



# Structural and functional benefits of *Ganoderma lucidum* polysaccharides due to space mutagenesis

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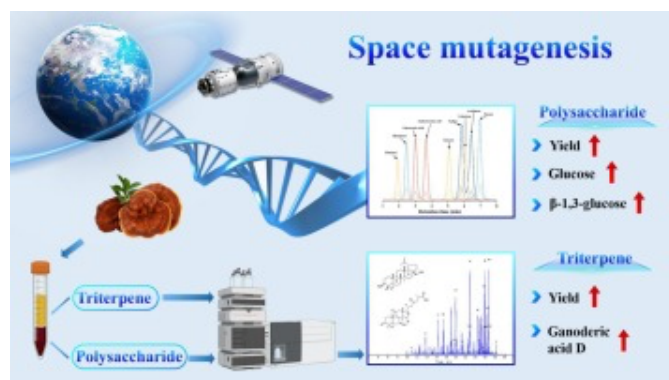
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## Abstract

*Ganoderma lucidum* samples from space-mutated strains transported by China's "Shenzhou" recoverable satellites, along with traditionally cultivated counterparts, were selected for this study. Polysaccharides from *Ganoderma lucidum* were extracted using a two-step sequential method integrating hot water and alkaline extractions. Liquid chromatography-mass spectrometry (LC-MS) was employed to characterize polysaccharide structures (monosaccharide composition and glycosidic linkages) and profile terpenoids, assessing the impact of space mutation. Results showed a significant increase ( $p < 0.05$ ) in polysaccharide and terpenoid contents, with polysaccharide yield rising from 178.59 % to 265.38 %. Space-mutated polysaccharides exhibited higher glucose content, altered monosaccharide composition, lower molecular weight, and an enriched  $\beta$ -1,3-glucan structure. They also demonstrated enhanced anti-inflammatory activity, underscoring the functional benefits of space-induced modifications. This study presents the first comprehensive analysis of structural alterations in *Ganoderma lucidum* polysaccharides induced by space mutation. It provides valuable scientific insights into the space breeding of medicinal plants and suggests that space mutation technology could serve as a promising strategy for developing functional foods from *Ganoderma lucidum*.

## Graphical abstract



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## Introduction

*Ganoderma lucidum*, one of the most renowned medicinal fungi in China, has been esteemed in classical herbal medicine since the Eastern Han Dynasty. Over the past three decades, extensive domestic and international research has substantiated *Ganoderma lucidum*'s potential to inhibit tumor growth, enhance immune function, slow aging, and protect the liver (Ahmad, 2020; El Sheikha, 2022; Seweryn, Ziała, & Gamian, 2021). Owing to its broad spectrum of pharmacological effects and minimal toxicity, *Ganoderma lucidum* is clinically applied as an adjunct therapy for various conditions, including tumors, bronchitis, and neurasthenia (Cör Andrejč, Knez, & Knez Marevci, 2022; Wu et al., 2024). It has gained increasing recognition as a valuable resource for developing health-promoting functional foods and traditional Chinese medicine formulations.

*Ganoderma lucidum* contains a diverse array of chemical constituents, among which polysaccharides and triterpenoids are the most prominent. Extensive research has been conducted on these bioactive compounds (Swallah et al., 2023). Analytical studies on *Ganoderma lucidum* polysaccharides have demonstrated that they primarily consist of monosaccharides such as glucose and galactose (Zheng et al., 2024). These polysaccharides play a crucial role in regulating blood glucose and lipid metabolism (Zhong et al., 2024). *Ganoderma lucidum* is also abundant in triterpenoids, particularly tetracyclic triterpenoids, with approximately 200 distinct types identified and characterized. Studies suggest that these triterpenoids play a significant role in lowering cholesterol levels and exerting other beneficial biological effects (Zheng et al., 2024).

In China, the functional components of *Ganoderma lucidum*, particularly polysaccharides, exhibit significant variation and instability, with cultivation restricted to specific regions (Ren, Zhang, Yu, & Zhang, 2020). The scarcity of high-quality strains remains a pressing challenge requiring immediate attention. Spaceflight mutagenesis, an innovative breeding technique, integrates elements of space

science, biology, and agricultural breeding (Iyer, Mhatre, Gilbert, & Bhattacharya, 2022). Studies have shown that space-related factors, including particle radiation, microgravity, and weak magnetic fields, can modify the physiological and biochemical characteristics of strains, resulting in the emergence of novel varieties (Chen et al., 2024). In recent years, China's frequent launches of recoverable satellites have underscored the significance of space breeding. The influence of space-related factors on plant growth and development has become a focal point of scientific inquiry. Space breeding of medicinal plants is a distinctive Chinese initiative. Since its inception in 1999, the Shenzhou spacecraft program has facilitated the transport of over 100 medicinal plant species for space breeding (Xia, Li, Liang, Zhang, & Yan, 2021). Compared to conventional breeding techniques, space breeding is distinguished by a higher frequency of beneficial mutations and greater genetic stability, promoting the development of high-quality varieties. Additionally, it generates unique mutational resources absent in nature, offering novel germplasm with diverse bioactive components (Jin et al., 2022).

To date, there has been no investigation into the variations of bioactive compounds in *Ganoderma lucidum* when comparing space-induced mutant strains with those cultivated through traditional methods. Therefore, in this study we employed space mutagenesis combined with liquid chromatography-mass spectrometry (LC-MS) to analyze the detailed structures of polysaccharides and triterpenoids. Notably, this is the first report elucidating the fine polysaccharide structure in space-mutated *Ganoderma lucidum*, providing a scientific basis for breeding high-quality strains. In addition, we evaluated the anti-inflammatory activity of the extracted polysaccharides to explore the functional implications of the space-induced structural modifications.

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### Materials

Four samples of *Ganoderma lucidum* fruiting bodies were collected from Longquan, Zhejiang Province, China. These included one space-mutated strain cultivated in a field greenhouse (short-log method with soil covering) and three non-mutated strains (one cultivated in a field greenhouse and two cultivated under natural forest conditions in different regions). To develop the space-mutated strain, high-performing *Ganoderma lucidum* cultures were sent to space aboard a Shenzhou

recoverable satellite ...

## Analysis of basic physicochemical properties

Polysaccharides and triterpenes are the primary bioactive components of *Ganoderma lucidum*. Hot water extraction is the most widely applied method for polysaccharide isolation, whereas acid and alkali extractions are commonly utilized to break down cell wall-bound polysaccharides (Zhong et al., 2024). Terpenes are typically extracted using organic solvents such as ethanol and methanol, followed by concentration and drying (Zhong et al., 2024). Recent advancements, including microwave, ...

## Conclusion

This study employed LC-MS and other analytical methods to compare the fine structures of *Ganoderma lucidum* polysaccharides from four different strains. Spaceflight significantly increased both polysaccharide and triterpenoid levels in *Ganoderma lucidum* compared to the control, and the space-mutated polysaccharides exhibited a distinctive  $\beta$ -1,3-glucan-rich structure. Moreover, polysaccharides from the space-mutated strain demonstrated stronger anti-inflammatory activity than those from the ...

## CRedit authorship contribution statement

**Bingqian Nie:** Writing – original draft, Data curation. **Jianle Chen:** Supervision. **Wenkang Wang:** Data curation. **Tong Cao:** Data curation, Writing – review & editing. **Xingqian Ye:** Supervision. **Shiguo Chen:** Supervision. **Zhiqiang Hou:** Writing – original draft, Supervision. ...

## Declaration of competing interest

The authors declare no conflict of interest. ...

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## References (53)

M.F. Ahmad

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Journal of Ethnopharmacology (2020)

A. Apriyanto *et al.*

[A review of starch, a unique biopolymer-structure, metabolism and in planta modifications](#)

Plant Science (2022)

X. Chen *et al.*

[Mapping and identification of a reverse mutation of \*Rht2\* that enhances plant height and thousand grain weight in an elite wheat mutant induced by spaceflight](#)

Plant Physiology and Biochemistry (2024)

T. Hong *et al.*

[Applications of infrared spectroscopy in polysaccharide structural analysis: Progress, challenge and perspective](#)

Food Chemistry: X (2021)

Z.Q. Hou *et al.*

[High pressure processing accelerated the release of RG-I pectic polysaccharides from citrus peel](#)

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Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy (2022)

C.Y. Liang *et al.*

[Review of the molecular mechanisms of \*Ganoderma lucidum\* triterpenoids: Ganoderic](#)

acids a, C2, D, F, DM, X and Y

European Journal of Medicinal Chemistry (2019)

W. Liu *et al.*

Characterization and antioxidant activity of two low-molecular-weight polysaccharides purified from the fruiting bodies of *Ganoderma lucidum*

International Journal of Biological Macromolecules (2010)



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## Cited by (6)

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